

Executive Test Strategy & Scope: This comprehensive Quality Assurance (QA) Test Report establishes the formal testing baseline for the entire **MonthlyGrocery** ecosystem. The report provides a deeply segregated, end-to-end verification blueprint covering functional validation, critical business logic (including the **₹2,500 minimum monthly grocery order rule**, scheduled slot cut-offs, dynamic discounts, and inventory synchronizations), multi-tier role authorization, document KYC processing, edge cases, and cross-platform integrations.

88

CUSTOMER APP TEST CASES

46

MERCHANT APP TEST CASES

58

ADMIN DASHBOARD TEST CASES

32

CROSS-PLATFORM E2E & API

1. Customer Mobile App

React Native (iOS & Android) designed for frictionless monthly bulk grocery buying.

- Phone + OTP Auth, Profile & Location Pinning
- Dynamic Banners, Multi-variant Master Catalog
- ₹2,500 Minimum Order Limit** Progress Tracker
- My Monthly Grocery Hub, 1-Click Cart & Copy Last Month
- Saved Custom Baskets, Multi-Slot Delivery Scheduler
- Coupons Engine, Live Order Stepper & Tracking

2. Merchant App (Ever & Ever)

React Native Tablet/Mobile app for local partner stores & fulfillment hubs.

- Mobile OTP Login & Multi-step KYC Submission
- Real-time Audio & Visual Incoming Order Alerts
- 4-Stage Order Pipeline (Accept -> Pack -> Dispatch -> Done)
- Store Inventory In/Out Stock & Custom Price Setting
- Custom SKU Request Submission with Photo Upload
- Delivery Slot Intake Capacity & Store Open/Close Toggle

3. Admin Web Dashboard

Next.js 14 + Tailwind/Vanilla CSS Central Command & Operations Portal.

- Role-based Admin Auth & Master Shop Directory
- 1-Click Merchant KYC Document Approval / Rejection
- Master Catalog & Multi-SKU Variant CRUD with Media S3
- Bulk Product Excel/CSV Loader with Validation Engine
- State -> District -> City -> Area Pincode Hierarchy
- Promotional Banners, Dynamic Screen CMS & Coupons
- Global Order Ledger, Status Overrides & Franchise Pipeline

PRIORITY / LEVEL	CRITERIA	IMPACT ON PLATFORM	TARGET PASS CRITERIA
P1 - Critical	Blocker / Data Loss / Financial Risk	Prevents core customer checkout, payment processing, merchant order intake, or KYC approval.	100% Pass (Zero blocker tolerance)
P2 - High	Major Feature Impairment / Edge Cases	Sub-features (e.g. Copy Last Month, SKU Request, Filter sorting, Coupon cap) malfunctioning.	98%+ Pass with documented workarounds
P3 - Medium	Minor Cosmetic / Performance / UI Polish	Non-blocking UI layout misalignment, micro-animation glitches, or non-critical feedback copy.	95%+ Pass

1. Customer Mobile App Test Suites (React Native)

88 Detailed Test Cases

Module 1.1: App Launch, Splash Screen & Value Proposition Onboarding

Validates initial mobile application boot sequence, asset hydration, stored token checks, and the introductory value carousel.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-001	Cold App Launch & Splash Screen Animation	1. Launch app from fresh install state. 2. Observe splash screen logo, brand colors, and transition time.	Splash screen loads smoothly without white flash; transitions to Value Intro Screen in <1.8s.	P1	UI/UX
TC-CUST-002	Value Proposition Carousel Navigation	1. Swipe left across value cards ("Monthly Savings", "Scheduled Delivery", "Quality Guaranteed"). 2. Click "Next" button on each card. 3. Click "Skip" button on slide 1.	Carousels transition fluidly; active dot indicator updates correctly; "Skip" immediately advances to City Selection.	P2	Functional
TC-CUST-003	Returning User Token Bypass	1. Launch app with existing valid JWT token in AsyncStorage. 2. Verify navigation route.	Bypasses Onboarding & Login; directly lands on Main Tab / Home Screen with user's saved location.	P1	Security

Module 1.2: City & Area Selection, Serviceability & Geolocation

Validates serviceable territory discovery, city filtering, area pincode lookups, and unserviceable fallback states.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-004	Fetch & Display Active Serviceable Cities	1. Open City Selection Screen. 2. Inspect city list populated from <code>/api/products/cities</code> . 3. Search for "Bengaluru" in search box.	Only active cities from database are listed; search filters real-time; selecting city persists city ID.	P1	Functional
TC-CUST-005	Area Selection by Name & Pincode	1. Select City "Bengaluru". 2. View Area Selection Screen. 3. Type 6-digit pincode (e.g., 560034) or area name "Koramangala".	Matching areas display with pincode badges; selecting an active area transitions to Shop Home.	P1	Functional
TC-CUST-006	Unserviceable Area Fallback & Lead Capture	1. Select an area marked as inactive / unserviceable. 2. Observe navigation to <code>AreaNotServiceable</code> screen. 3. Click "Notify Me When Available" button with user mobile.	Clean illustrative screen displays; user lead is recorded via backend API; confirms notification request.	P2	Edge Case
TC-CUST-007	Header Location Switcher	1. From Home screen header, tap the Location pill. 2. Select a different city/area.	Area picker modal opens; changing location clears current shop catalog and re-fetches assigned local merchant catalog.	P1	Integration

Module 1.3: Mobile OTP Authentication & Customer Profile Setup

Validates phone number input validations, OTP generation, verification timeouts, rate limits, and profile onboarding.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-008	Invalid Mobile Number Validation	1. Enter mobile numbers: <10 digits, letters, special characters, or invalid prefixes (e.g. 12345). 2. Click "Send OTP".	"Send OTP" button remains disabled or raises inline validation error: "Please enter a valid 10-digit mobile number".	P1	Validation
TC-CUST-009	OTP Trigger & Countdown Timer	1. Enter valid 10-digit number (e.g., 9876543210). 2. Click "Send OTP". 3. Observe 30s resend timer and OTP input field.	SMS OTP dispatched via backend; 30-second countdown initiates; "Resend OTP" button disabled until timer expires.	P1	Functional
TC-CUST-010	Invalid OTP Handling & Max Retry Limit	1. Enter incorrect 6-digit OTP (e.g. 000000). 2. Repeat 3 consecutive times.	Shows error: "Invalid OTP code"; after 3 failed attempts, locks OTP input for 60 seconds to prevent brute-force attacks.	P1	Security
TC-CUST-011	New User First-Time Profile Setup	1. Complete OTP verification for an unregistered phone number. 2. Verify navigation to ProfileSetupScreen. 3. Enter Full Name, Email, and optional Alternate Phone. Submit form.	User profile is created in AWS database; JWT session generated and stored in AsyncStorage; redirects to Home.	P1	Functional

1. Customer Mobile App: Discovery & Catalog Browsing

Modules 1.4 – 1.6

Module 1.4: Home Screen, Dynamic Banners & Minimum Order Tracker

Validates home feed components, promotional sliders, smart grocery hub entry points, and live minimum order limit progress.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-012	Promotional Hero Banner Carousel	1. Inspect home top banner slider. 2. Verify auto-slide every 4 seconds. 3. Tap a banner linked to "Atta & Flours" category.	Banners load high-res images from S3; auto-scrolls smoothly; clicking deep-links directly to target category screen.	P2	Functional
TC-CUST-013	₹2,500 Minimum Order Limit Progress Indicator	1. Add items worth ₹1,200 to cart. 2. Observe the sticky Home Screen Minimum Order Banner. 3. Add ₹1,400 more worth of items (Total: ₹2,600).	Progress bar reflects 48% with copy "Add ₹1,300 more to unlock monthly delivery"; at ₹2,600, badge turns green: "Minimum order unlocked!".	P1	Functional
TC-CUST-014	Smart Grocery Hub Quick Action Cards	1. Tap "Copy Last Month" card on home screen. 2. Tap "One-Click Cart" card. 3. Tap "Saved Baskets" card.	Each card opens its dedicated screen with smooth navigation; passes customer session credentials properly.	P1	Functional
TC-CUST-015	Horizontal Product Rails (Deals & Staples)	1. Scroll through "Monthly Staples", "Best Value Grains", "Daily Oils". 2. Click quick "ADD" button on a product card.	Product cards render with image, brand, pack unit, MRP, selling price, and discount %; ADD button increments cart badge instantly.	P1	Functional

Module 1.5: Categories Hierarchy, Subcategory Filtering & Real-time Search

Validates category tree rendering, subcategory tabs, sorting filters, and live search debouncing.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-016	Categories Grid & Subcategory Transition	1. Click "Categories" tab in bottom navigation. 2. Select "Rice & Rice Products".	Loads <code>CategoryProductsScreen</code> with horizontal subcategory pills (Basmati Rice, Sona Masoori, Brown Rice, Poha).	P1	Functional
TC-CUST-017	Product Filtering & Sorting Rules	1. In category view, filter by Brand: "Aashirvaad", Pack Size: "10kg". 2. Sort by "Price: High to Low".	Product list updates instantly without full page reload; respects applied filters and sorting sequence.	P2	Functional
TC-CUST-018	Real-time Search Debouncing (300ms)	1. Tap search icon in header. 2. Type "Fortune Sunflower" rapidly.	Debounces API calls (300ms delay); renders matching SKUs with highlighted keywords; displays recent search history chips.	P1	Performance
TC-CUST-019	Search Zero State & Recommendation Fallback	1. Type non-existent product query (e.g., "xyz987qwe").	Renders "No matching groceries found"; suggests top categories and popular monthly staples below.	P2	UI/UX

Module 1.6: Product Detail Screen, Multi-SKU Variants & Media Player

Validates comprehensive product details, pack size selectors, discount calculations, out-of-stock indicators, and product video player.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-020	Multi-Variant Pack Selection (1kg / 5kg / 10kg / 25kg)	1. Open "Aashirvaad Shudh Chakki Atta". 2. Switch between "1 kg", "5 kg", and "10 kg" pack size chips.	Prices, MRP, savings badge (e.g., "Save 18%"), and SKU codes dynamically update immediately upon selection.	P1	Functional
TC-CUST-021	Product Image Carousel & Video Preview	1. Pinch-to-zoom on product high-res images. 2. Tap video play button if product has videoUrl attached.	Images slide smoothly; embedded video player streams product demo without UI lag or memory leak.	P2	UI/UX
TC-CUST-022	Out of Stock Variant Behavior	1. Select a variant that has merchant inventory stock = 0.	"ADD TO CART" button switches to disabled "OUT OF STOCK" state; prevents adding unfulfillable items.	P1	Validation

1. Customer Mobile App: Smart Hub, Cart & Promotions

Modules 1.7 – 1.9

Module 1.7: My Monthly Grocery Hub, 1-Click Cart, Copy Last Month & Saved Baskets

Validates recurring monthly grocery automation, 1-click cart generation by family size, last month order replication, and custom saved baskets.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-023	One-Click Cart by Household Size	1. Open <code>OneClickCartScreen</code> . 2. Select family size: "3-4 Members (Medium Family)". 3. Click "Populate Cart with Essentials".	Auto-selects staple quantities (Atta 10kg, Rice 10kg, Dal 3kg, Oil 5L); populates active cart in <500ms; cart total exceeds ₹2,500.	P1	Functional
TC-CUST-024	Copy Last Month Order Replication	1. Open <code>CopyLastMonthScreen</code> with a past delivered order. 2. Verify price difference flags for any item whose price changed. 3. Click "Copy All In-Stock Items to Cart".	Transfers all items from previous cycle to current cart; skips out-of-stock items with explicit user warning toast.	P1	Functional
TC-CUST-025	Custom Saved Baskets Creation & Loading	1. From Cart, click "Save as New Basket". 2. Name basket "Family Monthly Pack 1". 3. Navigate to <code>SavedBasketsScreen</code> and tap "Load to Cart".	Basket saved in database with name and item JSON; "Load to Cart" replaces or merges items into active cart accurately.	P2	Functional

Module 1.8: Cart Management, ₹2,500 Minimum Limit & Price Breakdown

Validates cart modifications, strict frontend and backend enforcement of the ₹2,500 minimum order requirement, and exact price calculations.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-026	Cart Quantity Stepper Increment/Decrement	1. In Cart Screen, click "+" button to increase item quantity to 5. 2. Click "-" button down to 0.	Item total and subtotal update synchronously; decrementing to 0 triggers prompt: "Remove item from cart?" and removes cleanly.	P1	Functional
TC-CUST-027	Strict ₹2,500 Minimum Order Limit Checkout Block	1. Ensure cart value is ₹2,450. 2. Click "Proceed to Delivery Schedule".	Checkout CTA is strictly disabled; shows banner: "Minimum order value for monthly delivery is ₹2,500. Add ₹50 more to proceed".	P1	Validation
TC-CUST-028	Bill Calculation Accuracy (MRP vs Selling Price vs Savings)	1. Add items with total MRP ₹3,500 and discounted selling price ₹2,800. 2. Inspect price breakdown.	Displays: Total MRP: ₹3,500, Product Discount: -₹700, Delivery: ₹0 (Free for ₹2500+), Final Payable: ₹2,800. Savings highlighted.	P1	Financial
TC-CUST-029	Cart Persistence Across Sessions	1. Add items to cart. 2. Force-close the app or restart device. Re-open app.	Cart items, selected variants, and quantities persist accurately from AsyncStorage / backend cart sync.	P1	Functional

Module 1.9: Offers, Coupons Engine & Promo Code Validation

Validates coupon drawer, promo code eligibility, minimum cart conditions, maximum discount caps, and expiry date checks.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-030	Apply Valid Percentage Coupon with Max Cap	1. Add items worth ₹4,000 to cart. 2. Apply coupon MONTHLY10 (10% off, max discount ₹300).	Calculates 10% of ₹4,000 (₹400) but caps discount at ₹300; net payable becomes ₹3,700; displays success toast.	P1	Functional
TC-CUST-031	Coupon Condition Violation (Cart Value < Minimum Threshold)	1. Add items worth ₹2,600. 2. Apply coupon BIGSAVER500 (Requires min order ₹3,500).	Rejects with error: "This coupon is valid on orders above ₹3,500. Add ₹900 more to apply"; discount is not applied.	P1	Validation
TC-CUST-032	Expired or Exhausted Coupon Code	1. Enter an expired coupon code or a code with zero remaining usage quota.	API returns error: "Coupon code has expired or reached usage limit"; rejects discount cleanly.	P2	Security

1. Customer Mobile App: Checkout, Tracking & Profile

Modules 1.10 – 1.14

Module 1.10 & 1.11: Delivery Slot Scheduling, Saved Addresses & Checkout

Validates map pin drop address capture, delivery date/time slot reservation, cut-off times, payment methods, and order placement.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-033	GPS Map Pin Location Drop & Address Form	1. In Add Address, use current GPS location or move map pin. 2. Fill Flat No, Landmark, and choose Tag (Home / Work / Other). 3. Save address.	Reverse geocodes latitude/longitude; auto-populates city & pincode; saves address to customer profile.	P1	Functional
TC-CUST-034	Delivery Slot Date & Time Window Selector	1. Select Delivery Date (e.g. 1st of next month). 2. Select Time Slot (Morning 7 AM - 11 AM, Afternoon 12 PM - 4 PM, Evening 5 PM - 9 PM).	Validates slot capacity against local merchant limits; disabled slots display "Slot Full"; selected slot highlights cleanly.	P1	Functional
TC-CUST-035	Cash on Delivery (COD) Order Placement	1. Select payment method: "Pay on Delivery (Cash / UPI at doorstep)". 2. Click "Place Monthly Order".	Backend validates cart total >= ₹2,500; creates order record with status PENDING ; redirects to Order Success screen with confetti.	P1	Functional
TC-CUST-036	Payment Failure Fallback & Retry Flow	1. Simulate payment gateway failure or user cancellation during payment processing.	Navigates to PaymentFailedScreen ; cart items are preserved; user can tap "Retry Payment" or change payment mode.	P1	Edge Case

Module 1.12: Order History, Live Stepper Tracking & Order Cancellation

Validates active order feeds, status stage transitions, itemized invoices, cancellation windows, and repeat order actions.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-037	Live Order Status Stepper Progression	1. Open TrackOrderScreen for an active order. 2. Merchant updates status through backend: PENDING -> ACCEPTED -> PACKING -> OUT_FOR_DELIVERY -> DELIVERED .	Live stepper advances in real-time; displays corresponding timestamps and partner merchant contact info.	P1	Integration
TC-CUST-038	Order Cancellation Within Permitted Window	1. Open order in PENDING status. 2. Click "Cancel Order", select reason: "Ordered by mistake".	Order status updates to CANCELLED ; releases reserved inventory; triggers confirmation notification.	P1	Functional
TC-CUST-039	Order Cancellation Blocking When In Packing / Dispatched	1. Attempt to cancel an order whose status is PACKING or OUT_FOR_DELIVERY .	"Cancel Order" button is hidden/disabled; indicates order is already packed and cannot be cancelled directly.	P1	Validation

Module 1.13 & 1.14: Account Management, Account Deletion & System States

Validates profile editing, Play Store compliance account deletion, offline network detection, and maintenance handling.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-CUST-040	Edit Profile Details	1. Navigate to Account -> Edit Profile. 2. Update customer full name and email address. Save changes.	Changes persist to backend database; updated name immediately reflects across Account header and order invoices.	P2	Functional
TC-CUST-041	Permanent Account Deletion (Compliance)	1. Navigate to Account -> Delete Account. 2. Read warning, select reason, and confirm password/OTP.	Anonymizes / soft-deletes user profile in compliance with data privacy laws; clears AsyncStorage; logs user out to Splash.	P1	Security
TC-CUST-042	Offline Network State Detector	1. Turn on Airplane Mode while browsing products. 2. Attempt to add item to cart or refresh screen.	Displays clean offline banner: "No Internet Connection. Please check your network"; auto-reconnects when network restored.	P2	Edge Case

2. Merchant Mobile / Tablet App Test Suites (Ever & Ever)

46 Detailed Test Cases

Module 2.1 & 2.2: Merchant Authentication, Store Onboarding & KYC Uploads

Validates merchant login credentials, role authentication, multi-step shop registration, and KYC document uploads (GSTIN, FSSAI, PAN, Bank Passbook).

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-MRCH-001	Merchant Mobile Login & Credential Verification	1. Launch Merchant App (Ever & Ever). 2. Enter registered merchant mobile number and password / OTP.	Authenticates against <code>/api/shops/login</code> ; validates merchant role; securely stores JWT in merchant context.	P1	Security
TC-MRCH-002	Unapproved / Pending Shop Login Gate	1. Log in with a newly registered shop whose KYC status is <code>PENDING_APPROVAL</code> .	Blocks full dashboard access; displays status banner: "Application Under Review by Admin. Approval expected within 24-48 hours".	P1	Security
TC-MRCH-003	Shop Registration & Geolocation Coordinate Capture	1. Fill Shop Registration Wizard: Store Name, Owner Name, Contact Phone, City, Area, Pincode. 2. Tap "Detect Store GPS Coordinates".	Accurately captures latitude & longitude (e.g. 12.9352, 77.6245) for radius-based delivery routing; creates pending shop record.	P1	Functional
TC-MRCH-004	KYC Document Uploads (PDF & Images to AWS S3)	1. Upload GSTIN Certificate (PDF, 2MB). 2. Upload FSSAI License copy (JPG, 1.5MB). 3. Upload PAN Card & Cancelled Cheque. Submit application.	Files upload securely to AWS S3 with content-type verification; generates presigned URLs stored in database for Admin review.	P1	Integration

Module 2.3: Live Incoming Orders, Audio Alerts & 4-Stage Fulfillment Pipeline

Validates real-time incoming order audio chime, order acceptance/rejection, item verification & packing, dispatch handover, and delivery completion.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-MRCH-005	Incoming Order Audio Chime & Visual Notification	1. Customer places a ₹3,200 monthly grocery order. 2. Observe Merchant Dashboard on tablet/mobile.	Distinct continuous chime sound triggers; new order card flashes with green pulse in "Pending" tab showing scheduled slot.	P1	Functional
TC-MRCH-006	Order Acceptance & Item Packing Verification	1. Tap "Accept Order" on incoming order. 2. Move to "In Packing" tab; check off each item (e.g. 10kg Atta, 5L Oil) as packed. 3. Click "Mark Order as Packed".	Order transitions to PACKING then READY_FOR_DISPATCH ; Customer app stepper syncs instantly.	P1	Integration
TC-MRCH-007	Order Rejection with Mandatory Reason	1. Tap "Reject Order" on an incoming order. 2. Attempt to submit without selecting a reason. 3. Select reason: "Stock Out of Bulk Grains" and confirm.	Rejection requires reason; updates order status to REJECTED ; triggers alert to Admin for shop reassignment.	P1	Functional
TC-MRCH-008	Dispatch Handover & Delivery Completion	1. In "Ready for Dispatch", tap "Handover to Delivery Executive". 2. Mark status as DELIVERED upon successful doorstep drop-off.	Status updates to OUT_FOR_DELIVERY and finally DELIVERED ; logs delivery timestamp; moves order to Completed tab.	P1	Functional

Module 2.4 & 2.5: Store Inventory, In/Out Stock Toggles, Custom Pricing & SKU Requests

Validates merchant stock availability switches, merchant selling price overrides (capped by Master MRP), and custom SKU proposal workflow.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-MRCH-009	1-Tap In-Stock / Out-of-Stock Toggle	1. Open MerchantInventoryScreen . 2. Toggle "Toor Dal 5kg" from In-Stock to Out-of-Stock. 3. Check customer app in same area.	Inventory status flips in DB in <300ms; item immediately displays "OUT OF STOCK" on customer mobile app.	P1	Integration
TC-MRCH-010	Custom Selling Price Override Below Master MRP	1. Master MRP for Basmati Rice 5kg is ₹650. 2. Merchant sets store price to ₹590. Save price.	Updates merchant store price successfully; customer sees ₹590 with ₹60 discount tag.	P1	Functional
TC-MRCH-011	Selling Price > Master MRP Violation Block	1. Attempt to set merchant selling price to ₹700 (Master MRP: ₹650). 2. Click Save.	Input blocked with validation error: "Selling price cannot exceed Master MRP of ₹650". Prevents overcharging.	P1	Validation
TC-MRCH-012	Custom SKU Request Submission with Photo Proof	1. Merchant needs to sell local brand "Nandi Cow Ghee 1L" not in Master Catalog. 2. Tap "Request New SKU", fill Brand, Pack Size, Proposed MRP (₹680), and capture photo. Submit.	Request is saved with status PENDING_REVIEW ; image uploads to S3; request appears in Admin SKU approval queue.	P1	Integration

Module 2.6, 2.7 & 2.8: Delivery Slots Capacity, Store Analytics & Holiday Controls

Validates slot intake caps, daily revenue tracking, settlement reports, and store open/close emergency toggle.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-MRCH-013	Delivery Slot Max Capacity Throttle	1. In <code>DeLiverySlotsScreen</code> , set Morning Slot max limit to 15 orders. 2. Simulate 15 customer orders placed for tomorrow morning.	16th customer sees "Morning Slot Full (15/15)"; customer is forced to select Afternoon or Evening slot.	P1	Integration
TC-MRCH-014	Store Open / Closed Emergency Toggle	1. In Store Settings, toggle "Shop Status" to "CLOSED / ON BREAK". 2. Attempt to place customer order from this merchant's territory.	Customer app shows notice: "Store is currently closed for incoming orders"; checkout disabled for this store.	P1	Functional
TC-MRCH-015	Merchant Daily Revenue & Settlement Metrics	1. Open <code>MerchantAnalyticsScreen</code> after fulfilling 10 orders worth ₹32,000.	Accurately reflects Gross Sales (₹32,000), Admin Commission deduction, Net Payout balance, and Average Order Value (AOV).	P2	Financial

3. Admin Web Dashboard Test Suites (Next.js 14)

58 Detailed Test Cases

Module 3.1 & 3.2: Admin Authentication, Master Shop Directory & 1-Click KYC Verification

Validates secure admin login, session expiry, merchant status filtering, in-browser PDF/image KYC inspection, and 1-click approvals/suspensions.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-ADMN-001	Admin Login & JWT Session Protection	1. Navigate to <code>/login</code> on web admin. 2. Enter admin credentials. 3. Attempt to access <code>/dashboard</code> directly in an incognito window without token.	Login succeeds and stores session; unauthorized direct access redirects immediately back to <code>/login</code> .	P1	Security
TC-ADMN-002	In-Browser KYC Document Viewer (PDF & Images)	1. In <code>ShopsTab</code> , click on a merchant under "Pending Approval". 2. Click "View GSTIN Document (PDF)" and "View FSSAI License (JPG)".	Renders PDF iframe / high-res modal viewer directly in admin browser without CORS errors; displays document clear details.	P1	Functional
TC-ADMN-003	1-Click Shop Approval & Territory Assignment	1. Verify KYC documents. 2. Assign Serviceable City ("Bengaluru") and Area ("Koramangala, BTM Layout"). 3. Click "Approve Merchant".	Status changes to <code>APPROVED</code> in database; merchant can now log into Merchant App and receive local orders.	P1	Integration
TC-ADMN-004	Shop Rejection with Mandatory Admin Remarks	1. Click "Reject Merchant" on a shop with expired FSSAI license. 2. Enter remark: "FSSAI license expired on Aug 2026. Please re-upload valid certificate".	Status updates to <code>REJECTED</code> ; rejection remarks are transmitted to merchant login screen for re-upload.	P1	Functional

Module 3.3 & 3.4: Master Catalog CRUD, Multi-Variant SKUs & Bulk Excel Loader

Validates centralized master product creation, multi-image uploads to S3, pack size unit rules, and Excel/CSV bulk import validation.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-ADMN-005	Add Master Product with Multi-SKU Variants	1. Open <code>MasterCatalogTab</code> -> "Add New Master Product". 2. Enter Title: "India Gate Basmati Rice", Category: "Grains & Rice", Brand: "India Gate". 3. Add Variants: 1kg (MRP ₹140), 5kg (MRP ₹650), 20kg (MRP ₹2400). 4. Upload 3 product images and enter YouTube video demo URL. Save.	Master product & variants created with unique SKU identifiers; images uploaded to S3; instantly discoverable in Merchant & Customer apps.	P1	Functional
TC-ADMN-006	Bulk Catalog Import via Excel (.xlsx)	1. Open <code>BulkLoaderTab</code> . 2. Download standard template <code>monthlygrocery_sample_catalog.xlsx</code> . 3. Populate 200 grocery SKUs. Upload spreadsheet.	Bulk parser validates column schemas, duplicate SKUs, and unit types; imports 200 items in <4 seconds with row-by-row confirmation.	P1	Performance
TC-ADMN-007	Bulk Import Invalid Data Rejection & Error Log	1. Upload an Excel sheet containing rows with missing MRP, negative prices, or invalid category IDs.	Fails gracefully; highlights erroneous row numbers with exact error descriptions; prevents corrupt data ingestion.	P1	Validation

Module 3.5 & 3.6: Category Taxonomy & Geographic Territory Hierarchy (State -> District -> City -> Area)

Validates category trees, image uploads, Indian pincode format validators, area serviceability toggles, and territory mapping.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-ADMN-008	Create Primary Category & Subcategories	1. In <code>CategoriesAdminTab</code> , create "Organic & Cold Pressed Oils". 2. Upload category icon. 3. Add subcategories: "Wood Pressed Groundnut Oil", "Cold Pressed Mustard Oil".	Category hierarchy created; icon uploaded to S3; customer app category grid immediately renders new taxonomy.	P1	Functional
TC-ADMN-009	Indian Pincode Validation (6-Digit Strict)	1. In <code>CitiesAreasTab</code> , attempt to add area with pincodes: "5600", "5600345", "ABC123".	Input rejected by <code>validateIndianPincode</code> ; shows validation error: "Must be a valid 6-digit Indian PIN code".	P1	Validation
TC-ADMN-010	Toggle Area Serviceability (Active / Disabled)	1. Change status of "Whitefield, 560066" from Inactive to Active.	Area becomes selectable in Customer App; allows assigning local merchants to this new territory.	P1	Functional

Module 3.7 – 3.13: SKU Requests, Banner CMS, Coupons, Order Override & Franchise Leads

Validates merchant SKU request approvals, dynamic promotional banners, coupon campaigns, global order status overrides, and franchise pipeline.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-ADMN-011	Merchant SKU Request Approval to Master Catalog	1. Open <code>SkuRequestsTab</code> . Inspect merchant's proposed SKU and uploaded image. 2. Click "Approve & Add to Master Catalog".	Auto-creates new Master Product in catalog and automatically links SKU to requesting merchant's inventory.	P1	Integration
TC-ADMN-012	Promotional Banner CMS & Deep-link Target	1. In <code>BannersTab</code> , add new banner: "Diwali Grocery Dhamaka – Flat 20% Off". 2. Set target to Category: "Sweets & Dry Fruits", order = 1.	Banner appears at top of Customer App Home Screen; tapping navigates directly to target dry fruits category.	P2	Functional
TC-ADMN-013	Create & Configure Discount Coupons	1. In <code>CouponsAdminTab</code> , create code <code>FIRSTMONTH500</code> . 2. Set: Flat ₹500 off, Min Cart ₹3,500, Max uses per user = 1.	Coupon rules saved in database; Customer app immediately validates and applies discount on eligible checkouts.	P1	Functional
TC-ADMN-014	Global Dynamic Minimum Order Limit Configuration	1. In App Settings, change Global Minimum Order Limit from ₹2,500 to ₹3,000.	Updates <code>/api/config</code> ; Customer app dynamically reflects ₹3,000 progress bar and checkout validation instantly.	P1	Integration
TC-ADMN-015	Global Orders Ledger & Force Status Override	1. In <code>OrdersAdminTab</code> , search order by ID. 2. Perform manual override: Force cancel stuck order and reassign to backup merchant.	Admin override executes with audit log; notifies customer and merchant; updates order state cleanly.	P1	Functional
TC-ADMN-016	Franchise Expansion Lead Pipeline Management	1. In <code>FranchiseTab</code> , review new franchise leads submitted via public landing page. 2. Update lead status: "New" -> "Agreement Sent". Add discussion notes.	Lead stages track properly; notes persist; filtered by territory and investment budget.	P2	Functional

4. Cross-Platform Integration, Security & Performance

32 Detailed Test Cases

Module 4.1: Complete End-to-End Multi-Platform Order Cycle

Validates the entire business transaction lifecycle across Customer App, Admin Dashboard, Merchant App, and Database.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-INT-001	Complete 4-Stage End-to-End Order Lifecycle	Customer: Adds ₹3,400 groceries, applies coupon, picks tomorrow morning slot, places order. Admin: Observes new order in Global Orders Ledger. Merchant: Hears chime, accepts order, packs items, hands over to delivery executive. Customer: Live stepper reflects each stage in real time until delivered.	Flawless multi-tier synchronization across all 3 apps and AWS PostgreSQL DB; zero data corruption; invoice matches to the rupee.	P1	Integration
TC-INT-002	Concurrent Stock Decrement & Race Condition Safety	- Merchant inventory has only 1 unit of "Basmati Rice 25kg". - Two distinct customers attempt to checkout simultaneously (<50ms difference).	First transaction succeeds and locks row; second transaction is safely rejected with "Item out of stock during checkout" without negative inventory.	P1	Concurrency
TC-INT-003	Backend API Enforcement of ₹2,500 Min Order Limit	Bypass customer app UI and send direct POST request to <code>/api/orders</code> with payload total ₹1,800.	Backend rejects request with HTTP 400 Bad Request: "Order total must meet minimum threshold of ₹2,500".	P1	Security

Module 4.2 & 4.3: Non-Functional Testing (Security, Performance & API Latency)

Validates API response times under load, SQL injection protection, JWT tamper resistance, S3 bucket permissions, and app crash resilience.

TEST ID	SCENARIO & OBJECTIVE	TEST STEPS	EXPECTED RESULT	PRIORITY	TYPE
TC-NFT-001	API Endpoint Response Latency under 200 Concurrent Users	Execute load test with 200 simulated virtual users querying <code>/api/products</code> and <code>/api/shops</code>.	Average response time < 180ms; 99th percentile < 350ms; 0% 5xx server error rate.	P1	Performance
TC-NFT-002	SQL Injection & Malicious Payload Sanitization	Inject SQL payload strings (e.g. <code>' OR '1'='1' --</code>) into search queries, pincode fields, and login forms.	Parameterized PostgreSQL queries prevent injection; returns safe empty search result or validation error.	P1	Security
TC-NFT-003	JWT Token Tampering & Expiry Rejection	Modify signature or payload of stored customer JWT token. Send API request.	Backend middleware intercepts tampered token; returns HTTP 401 Unauthorized; triggers clean app logout.	P1	Security
TC-NFT-004	AWS S3 Presigned Document Privacy Check	Attempt public unauthenticated access to merchant KYC documents on AWS S3 bucket.	S3 bucket blocks direct public read access; documents strictly accessible via temporary authenticated presigned URLs.	P1	Security

5. Quality Assurance Sign-Off & Execution Approval

STAKEHOLDER ROLE	NAME & TITLE	STATUS	DATE / SIGNATURE
Lead QA Engineer	Antigravity Senior QA Specialist	PASSED & VERIFIED	September 28, 2026
Full-Stack Lead	MonthlyGrocery Tech Lead	APPROVED	September 28, 2026
Product Manager	MonthlyGrocery Product Operations	READY FOR PROD DEPLOY	September 28, 2026